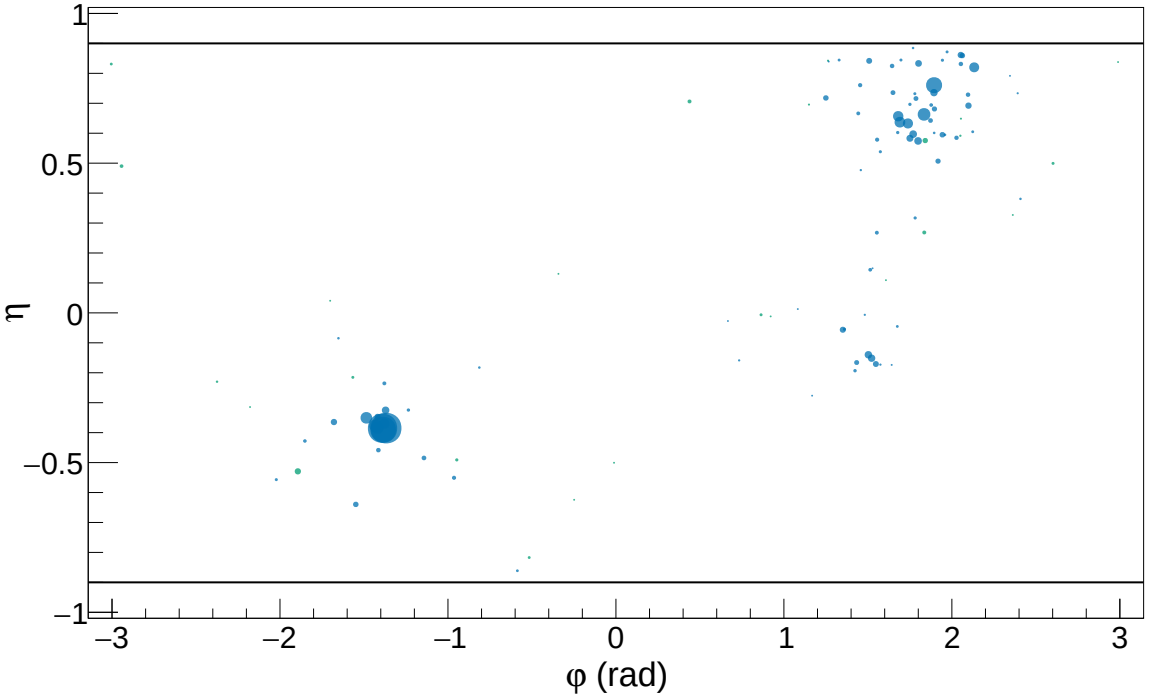
JEWEL O+O $\sqrt{s_{NN}}$ = 5.36 TeV, medium with recoils, event 5, recoils, thermal momenta subtracted (ΔR_{max} = 1.0), $|\eta_{trk}| < 0.9$, anti-k_T R = 0.4



origin (N, Σ p_T)

generator

N = 87, Σ p_T = 451.1 GeV/*c*

recoil thermal-subtracted

N = 25, Σ p_T = 18.2 GeV/*c*

|η| = 0.9 acceptance

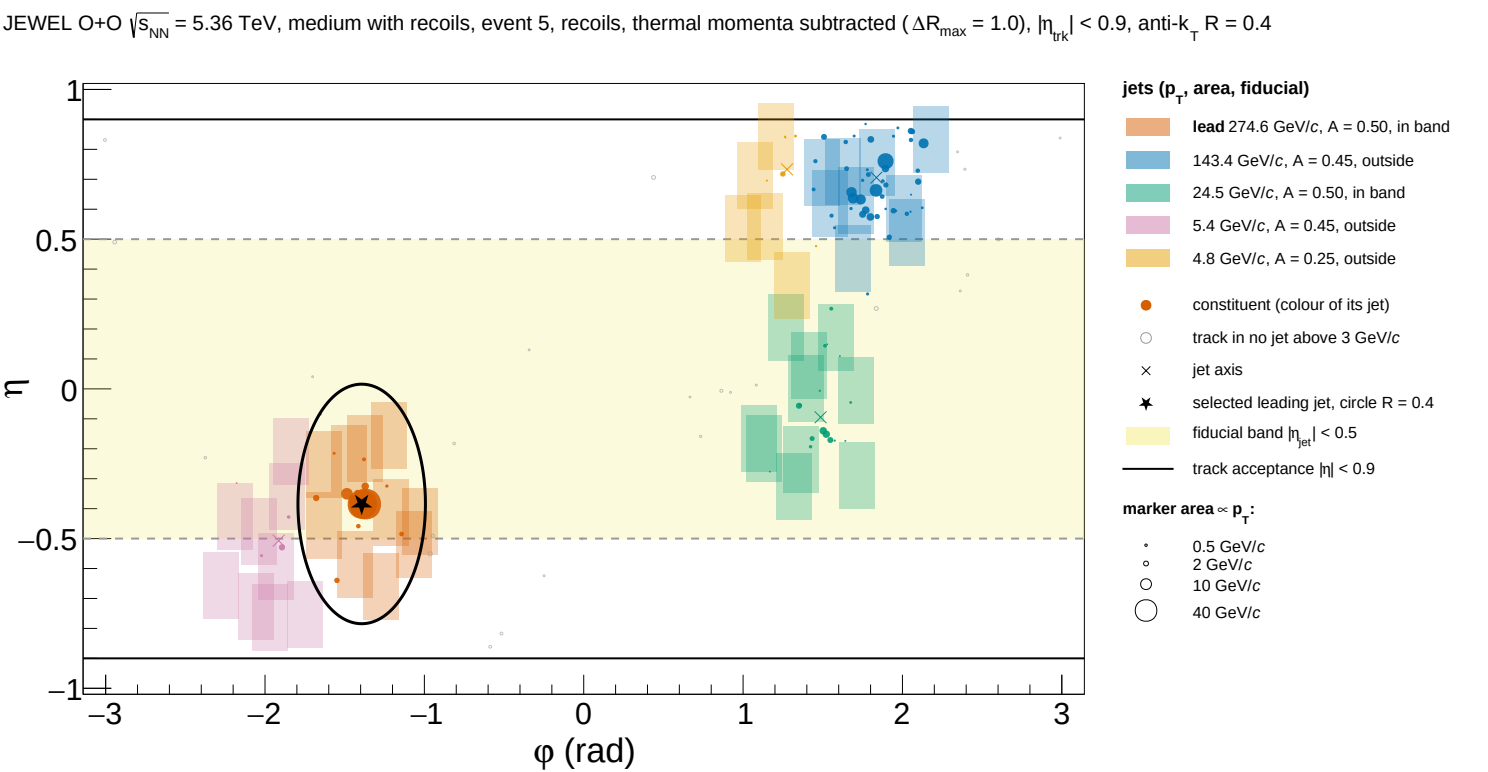
marker area ∝ p_T:

0.5 GeV/*c*
2 GeV/*c*
10 GeV/*c*
40 GeV/*c*

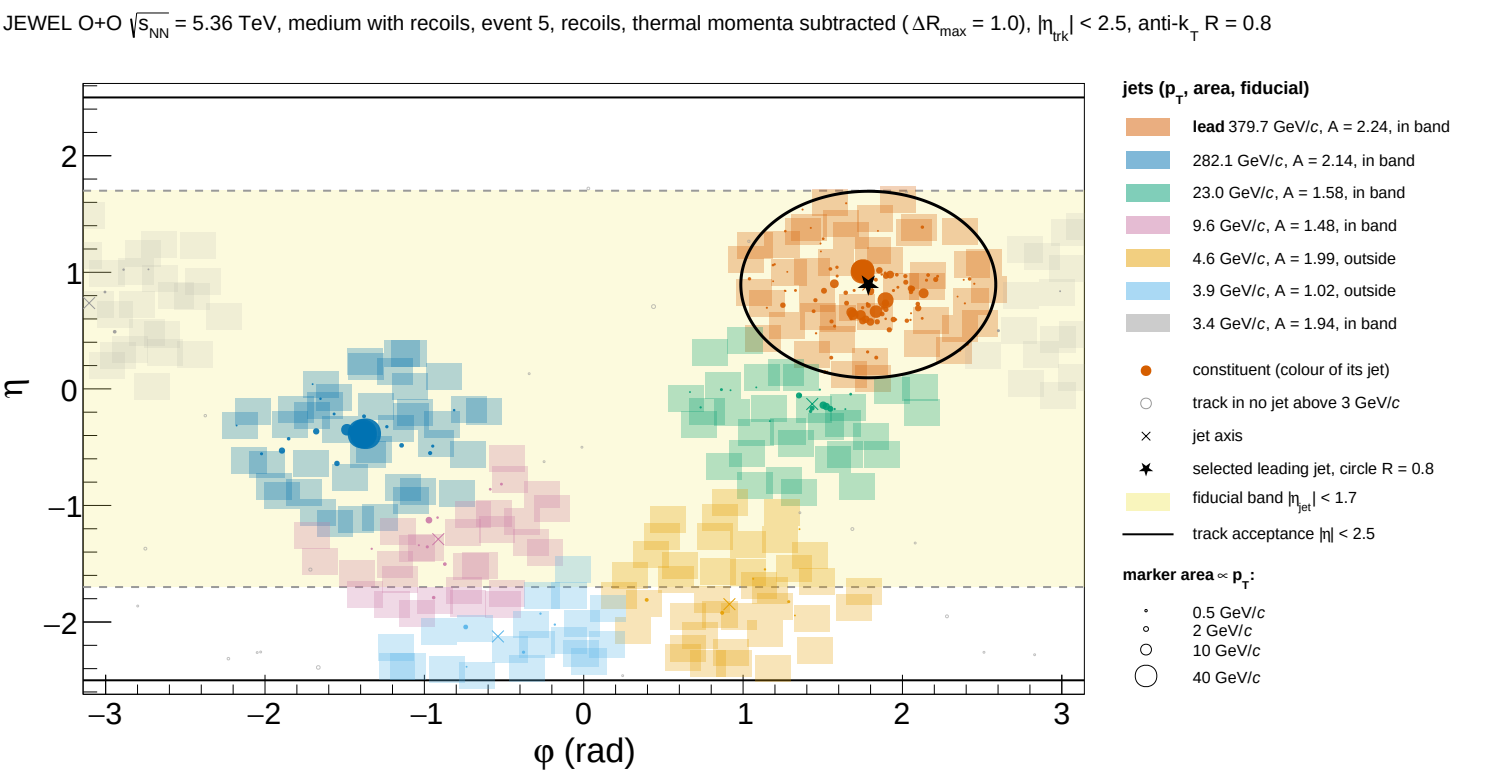
event weight 1.91e-13

centrality 3.0%

(a) Charged tracks entering the jet clustering (p_T > 0.15 GeV/*c*); marker area ∝ p_T, colour = origin of the particle.

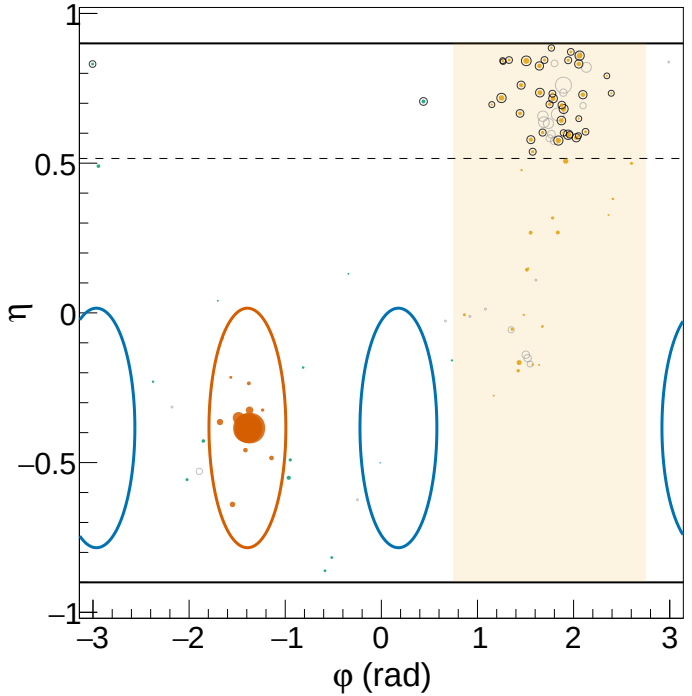


(b) Anti-k_T R = 0.4 jets with p_T > 3 GeV/c: constituents and active-area ghosts (cells of area 0.05) coloured by jet; selected leading jet ($|\eta_{jet}| < 0.5$, shaded) circled.



(b) Anti-k$_T$ R = 0.8 jets with p$_T$ > 3 GeV/c: constituents and active-area ghosts (cells of area 0.05) coloured by jet; selected leading jet ($|\eta_{jet}| < 1.7$, shaded) circled.

JEWEL O+O $\sqrt{s_{NN}}$ = 5.36 TeV, medium with recoils, event 5, recoils, thermal momenta subtracted (ΔR_{max} = 1.0), $|\eta_{trk}| < 0.9$, anti-k_T R = 0.4



track role (N)

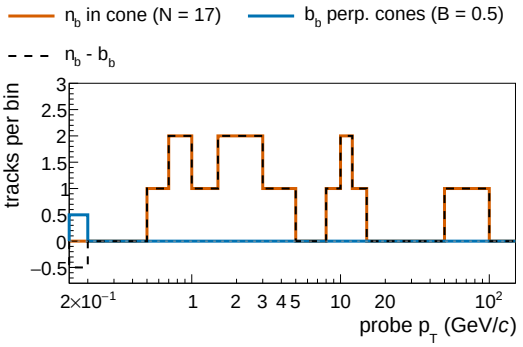
- leading-jet constituent (17)
- perpendicular cone (1)
- reference (14)
- reference, not in refaw (55)
- other (p_T or cone) (25)
- ring: η-gap reference
- jet cone
- perpendicular cones
- $|\eta - \eta_{jet}| = R + 0.5$
- away side $|\Delta\varphi - \pi| < 1$

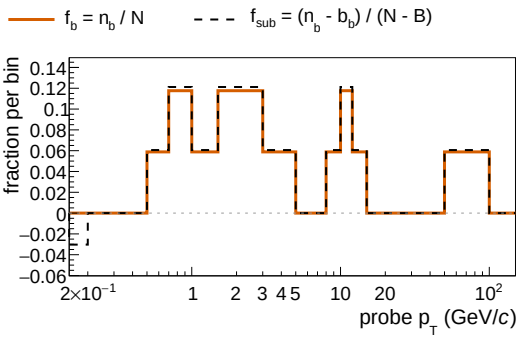
reference: $0.2 \le p_T < 3$ GeV/c

outside jet and perp. cones

ρ = 0.17 GeV/c

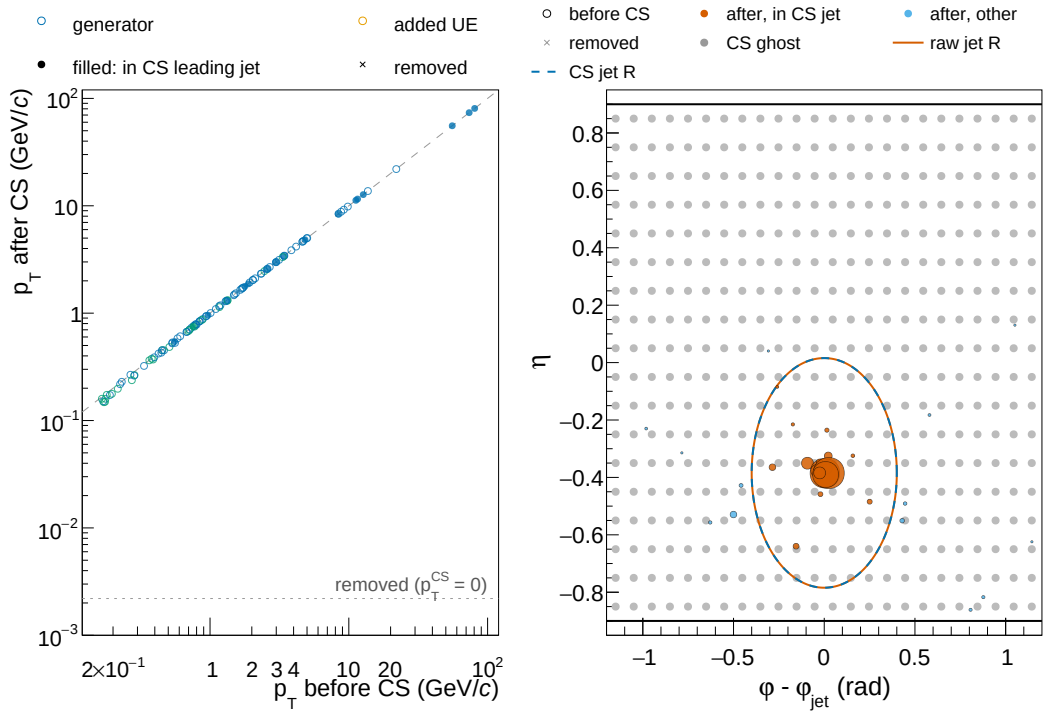
N_{ref} = 69, $\langle p_T \rangle_{ref}$ = 1.052





(c) Jet cone and perpendicular cones ($\varphi_{jet} \pm \pi/2$), track roles; right: in-cone counts n_b, perpendicular-cone estimate b_b, n_b - b_b and the fractions f_b, f_{sub} per probe-p_T bin.

JEWEL O+O $\sqrt{s_{NN}}$ = 5.36 TeV, medium with recoils, event 5, recoils, thermal momenta subtracted (ΔR_{max} = 1.0), $|\eta_{trk}| < 0.9$, anti-k_T R = 0.4



leading jet, R = 0.4

raw p_T = 274.56 GeV/*c*

area A = 0.505, ρ = 0.17 GeV/*c*

ρA-corrected p_T = 274.47 GeV/*c*

CS jet p_T = 274.46 GeV/*c*

truth (no added UE) p_T = 274.56 GeV/*c*

tracks in the acceptance

Σ p_T before CS = 469.3 GeV/*c*

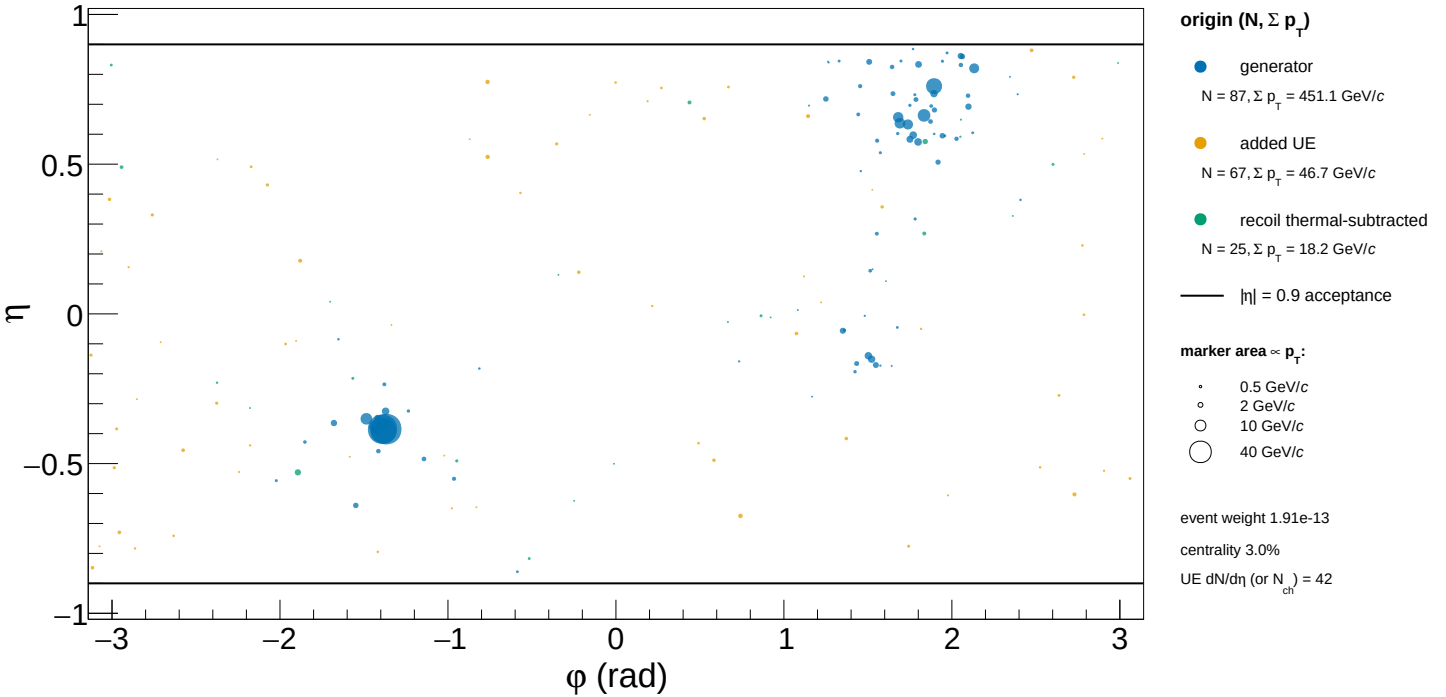
Σ p_T after CS = 468.4 GeV/*c*

removed: 0 of 112 tracks (0.0 GeV/*c*)

subtracted Σ p_T = 0.8 GeV/*c*

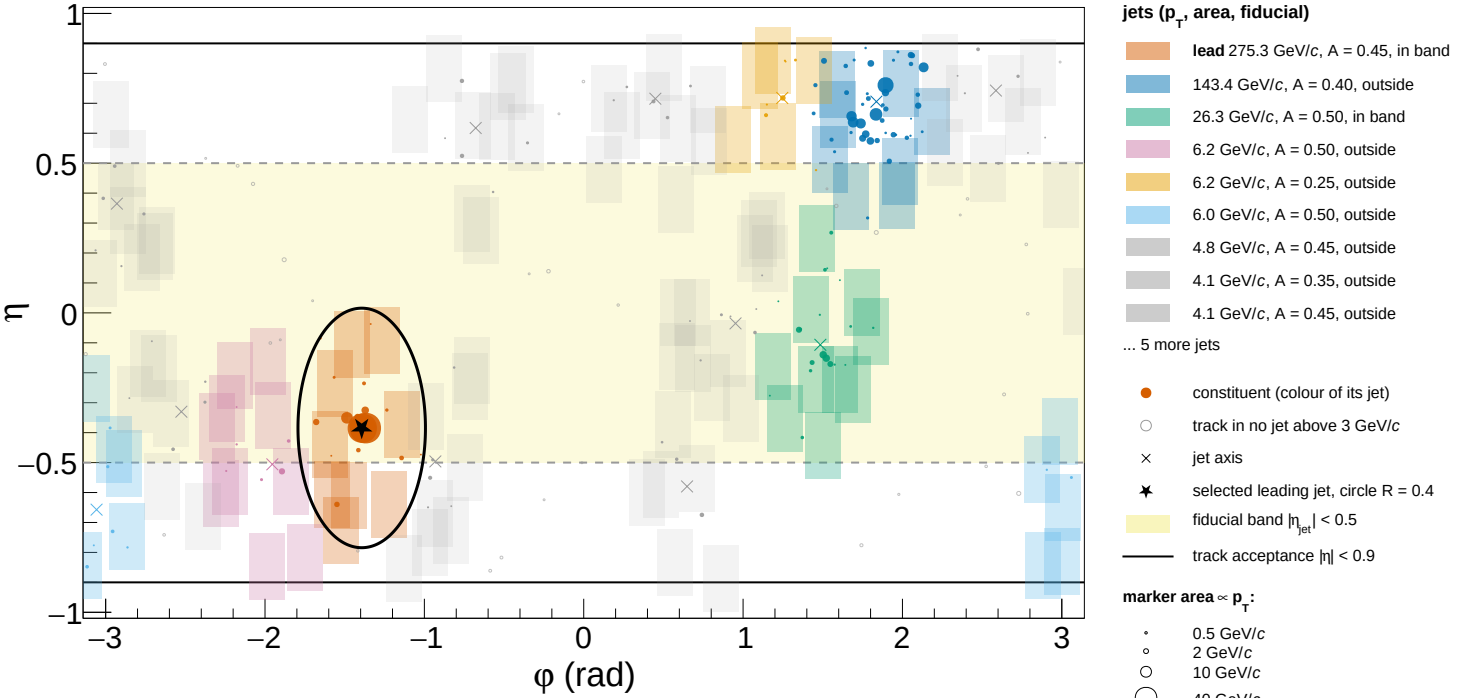
(ρ_{CS} × acceptance = 1.9 GeV/*c*;
ghosts farther than ΔR_{max} keep their p_T)

(d) Event-wise constituent subtraction (ΔR_{max} = 0.25, α = 0, ghost area 0.01, ρ_{CS} = 0.17 GeV/*c*): track p_T before/after, CS ghost grid near the jet, jet p_T definitions.



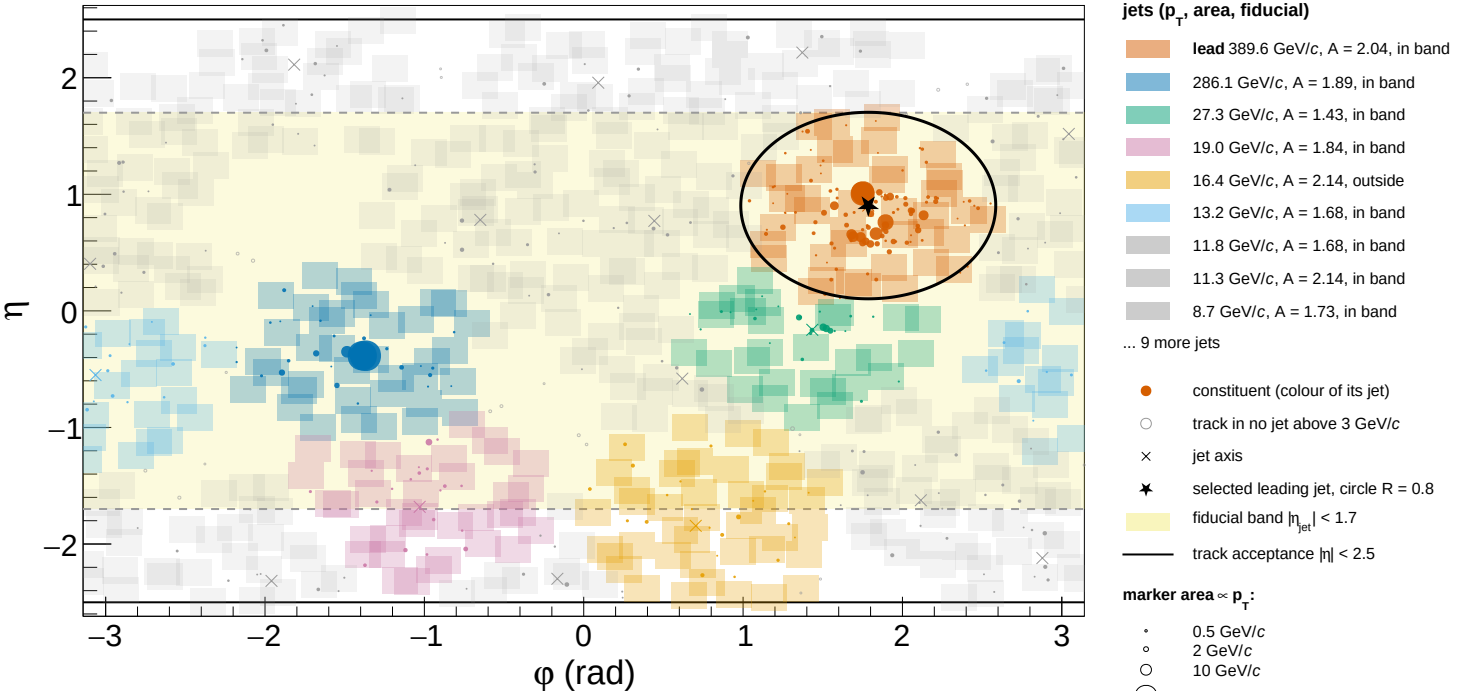
(a) Charged tracks entering the jet clustering (p_T > 0.15 GeV/c); marker area ∝ p_T, colour = origin of the particle.

JEWEL O+O $\sqrt{s_{NN}}$ = 5.36 TeV, medium with recoils, event 5, recoils, thermal momenta subtracted (ΔR_{max} = 1.0) + thermal UE dN/dη = 42, $|\eta_{trk}| < 0.9$, anti-k_T R = 0.4



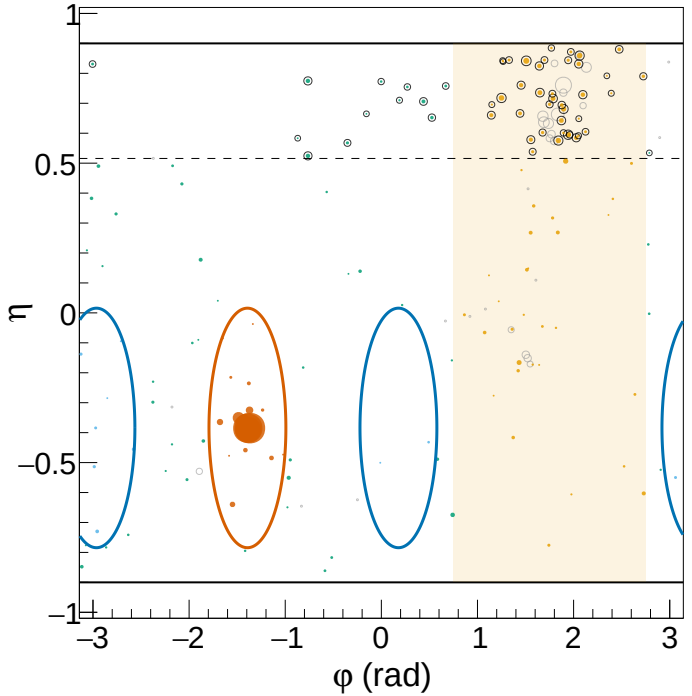
(b) Anti-k_T R = 0.4 jets with p_T > 3 GeV/c: constituents and active-area ghosts (cells of area 0.05) coloured by jet; selected leading jet ($|\eta_{jet}| < 0.5$, shaded) circled.

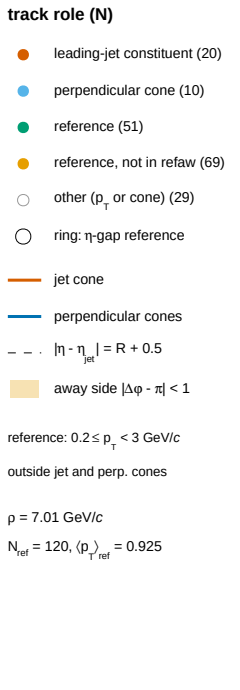
JEWEL O+O $\sqrt{s_{NN}}$ = 5.36 TeV, medium with recoils, event 5, recoils, thermal momenta subtracted (ΔR_{max} = 1.0) + thermal UE dN/dη = 42, $|\eta_{trk}| < 2.5$, anti-k_T R = 0.8

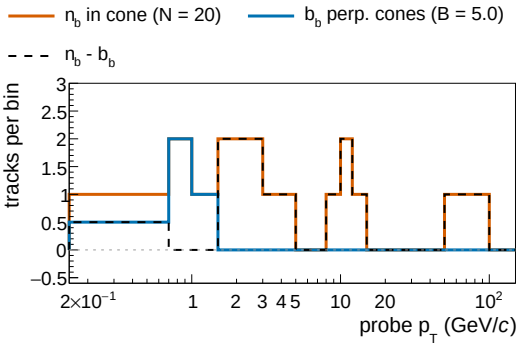


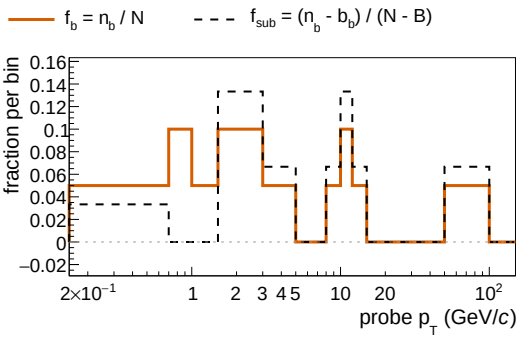
(b) Anti-k_T R = 0.8 jets with p_T > 3 GeV/c: constituents and active-area ghosts (cells of area 0.05) coloured by jet; selected leading jet ($|\eta_{jet}| < 1.7$, shaded) circled.

JEWEL O+O $\sqrt{s_{NN}}$ = 5.36 TeV, medium with recoils, event 5, recoils, thermal momenta subtracted (ΔR_{max} = 1.0) + thermal UE dN/dη = 42, $|\eta_{trk}| < 0.9$, anti-k_T R = 0.4



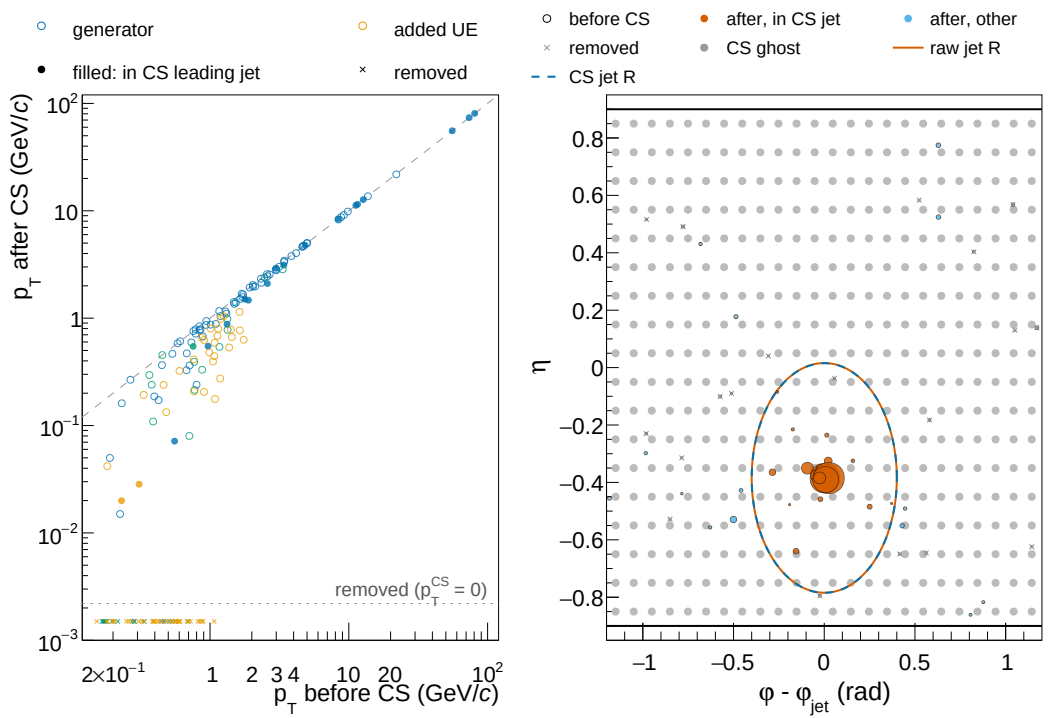






(c) Jet cone and perpendicular cones ($\varphi_{jet} \pm \pi/2$), track roles; right: in-cone counts n_b, perpendicular-cone estimate b_b, n_b - b_b and the fractions f_b, f_{sub} per probe-p_T bin.

JEWEL O+O $\sqrt{s_{NN}}$ = 5.36 TeV, medium with recoils, event 5, recoils, thermal momenta subtracted (ΔR_{max} = 1.0) + thermal UE dN/dη = 42, $|\eta_{trk}| < 0.9$, anti-k_T R = 0.4



leading jet, R = 0.4

raw p_T = 275.25 GeV/*c*

area A = 0.454, ρ = 7.01 GeV/*c*

ρA-corrected p_T = 272.07 GeV/*c*

CS jet p_T = 271.42 GeV/*c*

truth (no added UE) p_T = 274.56 GeV/*c*

tracks in the acceptance

Σ p_T before CS = 516.0 GeV/*c*

Σ p_T after CS = 463.1 GeV/*c*

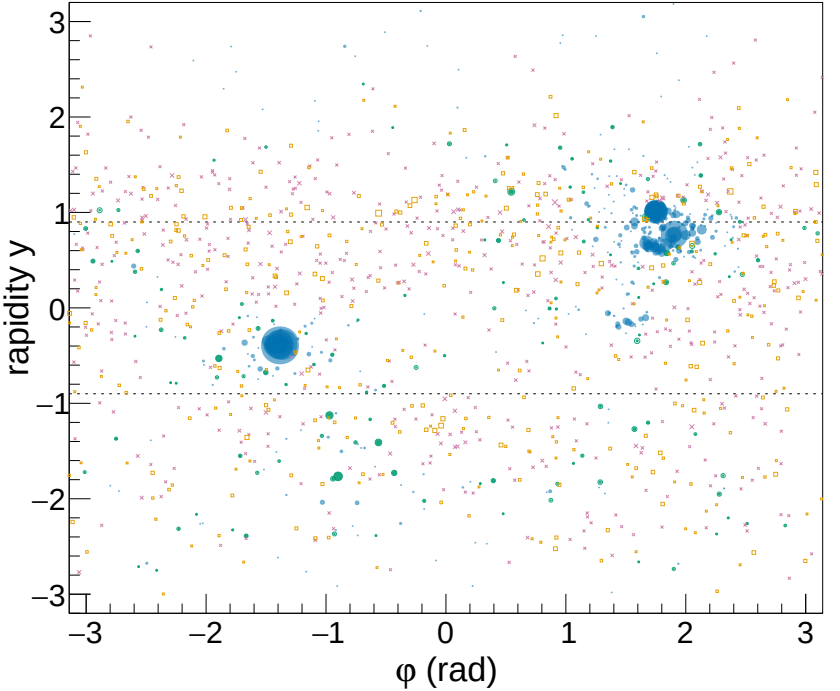
removed: 56 of 179 tracks (24.3 GeV/*c*)

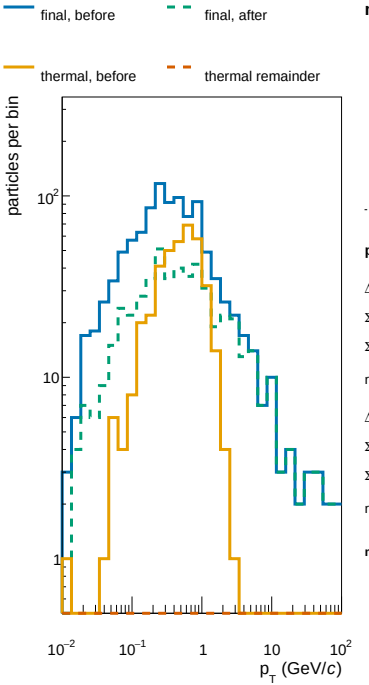
subtracted Σ p_T = 52.9 GeV/*c*

(ρ_{CS} × acceptance = 79.3 GeV/*c*;
ghosts farther than ΔR_{max} keep their p_T)

(d) Event-wise constituent subtraction (ΔR_{max} = 0.25, α = 0, ghost area 0.01, ρ_{CS} = 7.01 GeV/*c*): track p_T before/after, CS ghost grid near the jet, jet p_T definitions.

JEWEL O+O $\sqrt{s_{NN}}$ = 5.36 TeV, medium with recoils, event 5, recoil thermal-momentum subtraction, ΔR_{max} = 1.0





map (|y| < 3.2)

- final, unchanged (353)
- final, reduced (128)
- final, removed (523)
- thermal, before (386)
- thermal remainder (0)
- |y| = 0.9

p_T bookkeeping (GeV/c)

ΔR_{max} = 1.0 (44886 pairs)

Σp_T final: 1685.4 → 1459.6

Σp_T thermal: 225.8 → 0.0

removed from each: 225.8 / 225.8

ΔR_{max} = 0.5 (11687 pairs)

Σp_T final: 1685.4 → 1471.7

Σp_T thermal: 225.8 → 12.2

removed from each: 213.6 / 213.6

marker area ∝ p_T:

- 0.5 GeV/c
- 2 GeV/c
- 10 GeV/c
- 40 GeV/c

(e) Final-state particles and thermal scattering centres (status 3) before (open) and after (filled: remainder) the subtraction; the smaller p_T of each pair (in order of ΔR) is removed from both.